

percentage effect, since that ATT is dependent on the units in which the outcome is measured (Proposition 1).

In what follows, we consider alternative parameters that may be of interest when the marginal distributions of the potential outcomes are identified for some population of interest. Such identification is obtained in RCTs or under conditional unconfoundedness (for the full population), as well in instrumental variables settings (for the population of compliers), as these designs do not rely on functional form assumptions for identification. If the original identification strategy relies on a functional form assumption (e.g. parallel trends), then obtaining identification of the alternative parameters discussed below may require different identifying assumptions. We discuss these issues in detail in Section 5.2, where we revisit the difference-in-differences application in Sequeira (2016).

4.1 When the goal is interpretable units

We first consider the case where the researcher’s primary goal is to obtain a treatment effect parameter with easily interpretable units, such as percentages.

Normalizing the ATE in levels. One possibility is to target the parameter

$$\theta_{\text{ATE}\%} = \frac{E[Y(1) - Y(0)]}{E[Y(0)]},$$

which is the ATE *in levels* expressed as a *percentage of the control mean*. For example, if a researcher is studying a program D meant to reduce healthcare spending Y , then $\theta_{\text{ATE}\%}$ is the percentage reduction in costs from implementing the program. This parameter is point-identified and scale-invariant, and thus has an intuitive percentage interpretation. Importantly, however, $\theta_{\text{ATE}\%}$ is the percentage change in the average outcome between treatment and control, but is *not* an average of individual-level percentage changes.²² That is, $\theta_{\text{ATE}\%}$ does not take the form $E_P[g(Y(1), Y(0))]$, thus avoiding the trilemma in Proposition 2.

We note that $\theta_{\text{ATE}\%}$ is consistently estimable by Poisson regression (see [Gourieroux et al. \(1984\)](#); [Santos Silva and Tenreiro \(2006\)](#); [Wooldridge \(2010, Chapter 18.2\)](#)) under an appropriate identifying assumption. With a randomly assigned D , for example, estimation of $Y = \exp(\alpha + \beta D)U$ by Poisson quasi-maximum likelihood (QMLE) consistently estimates the population coefficient β , which satisfies $e^\beta - 1 = E[Y(1)]/E[Y(0)] - 1 = \theta_{\text{ATE}\%}$. In Section 5 below, we illustrate how $\theta_{\text{ATE}\%}$ can be estimated by Poisson regression in practice in several empirical examples, including both an RCT and DiD setting.

We also emphasize that $\theta_{\text{ATE}\%}$ is influenced by treatment effects along both the intensive and extensive margins. In particular, the numerator of $\theta_{\text{ATE}\%}$ is the ATE in levels. Thus, if an individual has a treatment effect of say 1, that contributes the same to $\theta_{\text{ATE}\%}$ regardless of whether their outcome changes from 0 to 1 (an extensive margin change) or 1 to 2 (an intensive margin change).

²²This is roughly analogous to how quantile treatment effects show changes in the quantiles of the potential outcomes distributions, but *not* the quantiles of the treatment effects (without further assumptions).

The parameter $\theta_{\text{ATE}\%}$ may therefore be attractive in settings where the researcher does not want to distinguish between the intensive and extensive margins. For example, if Y is a count of publications by a researcher in a particular year, and publications are sometimes zero owing to the idiosyncracies of the publication process, then it may be reasonable to view a change between 0 and 1 as similar to a change between 1 and 2. On the other hand, in settings where a zero corresponds to a distinct economic choice, such as not participating in the labor market, then it may be of interest to separate the effects along the intensive and extensive margin, as we discuss in more detail in [Section 4.3](#) below.

It is also worth noting that if the researcher has determined that the ATE in levels is not of economic interest, then similar issues will likely arise for $\theta_{\text{ATE}\%}$, since $\theta_{\text{ATE}\%}$ is just a re-scaling of the ATE in levels. For one, the ATE in levels (and hence $\theta_{\text{ATE}\%}$) imposes no diminishing returns, and thus might be dominated by individuals in the tail of the outcome distribution, particularly when the outcome is skewed. Whether this is warranted will depend on the economic question: if the policy-maker’s goal is to reduce healthcare spending, it may not matter whether the savings are produced mainly by reducing spending for a small fraction of individuals with catastrophic medical spending. On the other hand, a policy that increases every American’s income by \$100 and one that increases Elon Musk’s income by \$35 billion and has no effect on anyone else would have approximately the same value of $\theta_{\text{ATE}\%}$, yet the former may be vastly preferred by an inequality-minded policy-maker. We therefore next turn to alternative approaches that place less weight on the tails of the outcome distribution.

Normalizing other functionals. While $\theta_{\text{ATE}\%}$ normalizes the ATE by the control mean, one can obtain scale-invariance by normalizing other functionals of the potential outcomes distributions.²³ For example,

$$\theta_{\text{Median}\%} = \frac{\text{Median}(Y(1)) - \text{Median}(Y(0))}{\text{Median}(Y(0))},$$

is the quantile treatment effect at the median normalized by the median of $Y(0)$.²⁴ Put otherwise, it captures the percentage change in the median between the treated and control distributions. ($\theta_{\text{Median}\%}$ thus may be particularly relevant for politicians interested in maximizing the happiness of the median voter!) As is typically the case with quantile treatment effects, however, the numerator of $\theta_{\text{Median}\%}$ need not correspond to the median of individual-level treatment effects. Moreover, in many settings, decision-makers may care about treatment effects throughout the distribution, not just at the median, in which case $\theta_{\text{Median}\%}$ may not be the most economically-relevant parameter.

Normalizing the outcome. A related approach to obtaining a treatment effect with more intuitive units is to estimate the ATE for a transformed outcome that has a percentage interpretation. One example is to consider an outcome of the form $\tilde{Y} = Y/X$, where Y is the original outcome and X is some pre-determined characteristic. For example, suppose Y is employment in a particular

²³Indeed, any functional $\phi(P)$ is homogeneous of degree zero if and only if it can be written as the ratio of two homogeneous of degree one functionals.

²⁴Note that $\theta_{\text{Median}\%}$ is well-defined only if $\text{Median}(Y(0)) > 0$.

area. The treatment effect in levels for Y may be difficult to interpret, since a change in employment of 1,000 means something very different in New York City versus a small rural town. However, if X is the area’s population, then $\tilde{Y}$ is the employment-to-population ratio, which may be more comparable across places, and is already in percentage (i.e. per capita) units. We note that the ATE for $\tilde{Y}$ is a scale-invariant, point-identified parameter of the form $\theta = E_P[g(Y(1), Y(0), X)]$, and thus escapes the trilemma in [Proposition 2](#) by avoiding property (a).²⁵ The viability of this approach, of course, depends on having a variable X such that the normalized outcome $\tilde{Y}$ is of economic interest. We suspect that in many contexts, reasonable options will be available, including pre-treatment observations of the outcome (assuming these are positive), or the *predicted* control outcome given some observable characteristics (i.e., $X = E[Y(0) | W]$, for observable characteristics W).

A second example is to use $\tilde{Y} = F_{Y^*}(Y)$, where F_{Y^*} is the cumulative distribution function (CDF) of some reference random variable Y^* , as suggested in [Delius and Sterck \(2020\)](#). The transformed outcome $\tilde{Y}$ then corresponds to the rank (i.e. percentile) of an individual in the reference distribution, and the ATE for $\tilde{Y}$ can be interpreted as the average change in rank caused by the treatment. The ATE for $\tilde{Y}$ is unit-invariant so long as Y and Y^* are measured in the same units. Outcomes of this form have become increasingly popular in the literature on intergenerational mobility, where $\tilde{Y}$ corresponds to a child’s rank in the national income distribution. This approach has been found to yield more stable estimates than approaches using $\log(c + Y)$, which [Chetty et al. \(2014\)](#) show are sensitive to the choice of c .²⁶

Finally, the researcher might report treatment effects on transformed outcomes of the form $1[Y \geq y]$ for different values of y . For example, the researcher might report the impact of the treatment on the probability that an individual earns at least \$50,000, \$60,000, etc., and interpret it as the treatment effect on the probability of obtaining a “well-paying job.”²⁷ Such treatment effects have interpretable units as percentage points (i.e. changes in probabilities). We note that treatment effects for outcomes of this form combine the effect of the treatment along the intensive and extensive margin, since for example, a worker who has $Y(1) > \$50,000 > Y(0)$ could either not work under control ($Y(0) = 0$) or work under control but have earnings below \$50,000.

4.2 When the goal is to capture decreasing returns

We next consider the case where the researcher wants to capture some form of decreasing marginal utility over the outcome. For example, when Y is strictly positively valued, the ATE in logs corresponds to the change in utility from implementing the treatment for a utilitarian social planner with log utility over the outcome, $U = E[\log(Y)]$. Intuitively, this social welfare function captures the fact that the planner values a percentage point of change in the outcome equally for all individuals, regardless of their initial level of the outcome.

²⁵It is scale-invariant in the sense that $g(y_1, y_0, x) = g(ay_1, ay_0, ax)$.

²⁶Similar to the discussion in [Footnote 21](#), the treatment effect in ranks cannot be converted back to obtain the ATE in levels without additional assumptions.

²⁷The researcher could also report the implied CDF of $Y(1)$ and $Y(0)$, from which one can infer the treatment effect on outcomes of this form for all y .

Of course, log utility is not well-defined when there is an extensive margin: A coherent utility function defined with zero-valued outcomes must take a stand on the relative importance of the intensive versus extensive margins. Recall from [Section 2.1](#) that when using transformations like $\log(1 + y)$ or $\operatorname{arcsinh}(y)$, the scaling of the outcome implicitly determines the weights placed on these margins.

Instead of implicitly weighting the margins via the scaling of Y , a more transparent approach is to explicitly take a stand on how much one values the two margins of treatment. Of course, if one knows that their utility is captured by $U = E[m(Y)]$ (for a particular unit of Y , say earnings in dollars), then the ATE for $m(Y)$ is appropriate. If one is unsure exactly of their utility function, then a rough calibration is to specify how much one values a change in earnings from 0 to 1 relative to a percentage change in earnings for those with non-zero earnings. If, for example, one values the extensive margin effect of moving from 0 to 1 the same as a $100x$ percent increase in earnings, then one might consider setting $m(y) = \log(y)$ for $y > 0$ and $m(0) = -x$. The ATE for this transformation can be interpreted as an approximate percentage (log point) effect, where an increase from 0 to 1 is valued at $100x$ log points.²⁸

We emphasize that for a fixed value of x , this approach necessarily depends on the scaling of the outcome (thus avoiding the trilemma in [Proposition 2](#)). However, this may not be so concerning since the appropriate choice of x also depends on the units of the outcome—e.g., saying a change from 0 to 1 is worth $100x$ percent means something very different if 1 corresponds with one dollar versus a million dollars. In other words, ATEs for transformations such as $\operatorname{arcsinh}(Y)$ may be difficult to interpret because the scaling of the outcome implicitly determines the relative importance of the intensive and extensive margins; this approach avoids that difficulty by *explicitly* taking a stand on the tradeoff between these two margins. Nevertheless, a challenge with this approach is that researchers may have differing opinions over the appropriate choice of x (or more generally, over the appropriate utility function).

4.3 When the goal is to understand intensive and extensive margins

Finally, we consider the case where the researcher is interested in understanding the intensive and extensive margin effects separately. A common question in the literature on job training programs ([Card et al., 2010](#)), for instance, is whether a program raises participants’ earnings by helping them find a job—which would be expected only to have an extensive-margin effect—or by increasing human capital, which would be expected to also affect the intensive margin. In such settings, it is natural to target separate parameters for the intensive and extensive margins.

For example, the parameter

$$\theta_{\text{Intensive}} = E[\log(Y(1)) - \log(Y(0)) \mid Y(1) > 0, Y(0) > 0]$$

²⁸Note that this transformation will generally only be sensible if the support of Y excludes $(0, e^{-x})$, since otherwise the function $m(y)$ is not monotone in y over the support of Y . It is common, however, to have a lower-bound on non-zero values of the outcome; e.g., a firm cannot have between 0 and 1 employees. In our application to [Sequeira \(2016\)](#) below, we normalize the minimum non-zero value of Y to 1 when applying this approach.